SUPPORTING ADULT GUIDANCE

Parallel stories of change

Straight Lines and Connected Representations. For any supporting adult — teacher, practitioner, parent or carer. **You do not need to be a mathematics specialist to support this lesson.**

What this lesson does

The learner compares lines and recognises when they are **parallel**: lines are parallel exactly when they have the **same gradient m** and (usually) **different intercepts c**. The emphasis is on judging parallelism from the **gradient alone**, and on the idea that parallel lines **never meet**.

Driving Question: *Can two lines change alike but begin differently?*

The mathematics, in plain terms

Two straight lines are **parallel** when they run in the same direction and never meet, staying the same distance apart. The test is simple and it is all about the **gradient**: lines are parallel exactly when their **gradients are equal**. For $y = 2x + 1$ and $y = 2x + 5$, both climb 2 for every 1 across, so they are parallel; the different intercepts (1 and 5) just place one above the other. The classic trap is to think that lines crossing the y-axis at the **same** point are parallel — they are not: $y = 2x + 3$ and $y = 3x + 3$ share the intercept 3 but have different gradients, so they cross at (0, 3) and spread apart. Only equal gradients make lines parallel. Because parallel lines never meet, they matter in real life whenever two options change at the **same rate**: two phone plans at the same price per gigabyte, two gyms with the same per-session charge. Their cost lines are parallel, so the cheaper one (the smaller fixed fee) stays cheaper **forever** — the gap never closes and there is no break-even point. When the rates **differ**, though, the lines do cross — and that crossing point is exactly what the next lesson is about.

It is all in the gradient

The single idea to protect here is that **parallel depends on the gradient and nothing else**. Equal gradients mean parallel; unequal gradients mean the lines eventually cross, whatever their intercepts. Naming the common trap out loud — that a shared intercept does not make lines parallel — heads off a persistent error, and a quick example (two lines through the same point on the y-axis, spreading apart) settles it fast. It helps to keep 'parallel' tied to something physical: lines that never meet, or two deals that keep the same fixed gap forever. That real-world grip — same rate, so the cheaper option stays cheaper — makes the mathematics meaningful and sets up the next lesson, where lines with different gradients finally meet at a single crossing point.

The shape of the lesson

Timings are invitations, not deadlines. A learner may need much longer on one phase and skim another — that is expected. Each Cornerstone is given an honest mathematical role here:

Cornerstone	Its role here	Invitational timing
Connection	Parallel Line Challenge: pick the equation with the same gradient as the reference line.	~12 min

Movement	Make it parallel: match the reference's gradient with a different intercept until the lines run parallel.	~12 min
Reflection	Parallel or not?: judge pairs of lines by their gradients alone, spotting the shared-intercept trap.	~10 min
Creativity	Two plans, side by side: two deals at the same rate give parallel lines — which is always cheaper?	~6 min
Rest	A pause after seeing two lines change alike yet begin apart, running side by side forever.	~3 min
Nutrition	Contextual (Mode A): deals that charge the same rate are parallel — the cheaper one wins every time.	~2 min

Common misconceptions, and gentle repairs

These are the sticking points to expect. In every case, the aim is a question that returns the thinking to the learner — never to supply the answer.

Thinking a shared y-intercept makes lines parallel

Repair: “Do these two lines have the same gradient? If not, they cross — sharing an intercept does not make them parallel.”

Thinking lines must look identical to be parallel

Repair: “Parallel lines have the same steepness but usually a different intercept — same m, different c.”

Believing parallel lines eventually meet 'far away'

Repair: “If they have the same gradient, the gap never changes — they never meet, however far you go.”

Comparing intercepts instead of gradients to judge parallelism

Repair: “Which number decides steepness? Compare the gradients, not the intercepts.”

Assuming two real plans must cross at some point

Repair: “If both charge the same rate, the lines are parallel — there is no break-even; one is cheaper forever.”

Protecting the support pathway

The lesson offers support in a deliberate order: **Socratic prompt** → **first try** → **hint** → **second try** → **worked solution**. The worked solution for each question stays locked until the learner has used the prompt, used the hint, and checked two answers. This is by design: the steps *are* the learning.

Please hold this line

Do not read the worked solution aloud, or talk a learner straight to the answer, while it is still locked. If they are stuck, use the Reconnection Routes, the prompt and the hint, or ask one of the repair questions above. Arriving at a correct line of reasoning — even slowly — is worth far more than being handed the result.

Reconnection Routes

If an earlier idea is blocking progress, the lesson’s Reconnection Routes offer short recaps of exactly the prerequisites this lesson needs: reading the gradient and the intercept from an equation, and knowing that lines are equally steep when their gradients match. A learner can choose one independently, or you can invite one without requiring it. Using a route is part of learning, not evidence of falling behind, and it always returns the learner to their place.

Supporting through questions: an example

Keep the attention on the gradient:

- “What is the gradient of each line? Are they the same?”
- “These cross the y-axis at the same point — but are their gradients equal?”
- “Write a line parallel to this one. What must stay the same, and what can change?”
- “Both plans charge the same per unit — so are their cost lines parallel? Which is cheaper, and does that ever change?”

If they stall, that is the moment for a Reconnection Route or the lesson’s prompt and hint — not for the answer.

Questions you might have

“How can I tell instantly if two lines are parallel?”

Compare their gradients. If the number in front of x is the same, they are parallel — the intercepts do not matter for this. Different gradients mean they will cross somewhere.

“Can parallel lines ever meet?”

No. Equal gradients means the vertical gap between them never changes, so however far you extend them they stay the same distance apart and never touch.

“Why does this matter in the real world?”

When two deals charge the same rate, their cost lines are parallel, so the one with the smaller fixed fee is cheaper forever — there is no crossover to hunt for. When the rates differ, the lines cross, and finding that crossing point is the next lesson.

Keeping things regulated and calm

- Keep to one clear step at a time; there is no time pressure and no benefit to speed.
- Treat a wrong or unfinished answer as information for the next step, never as failure.
- Let the learner choose how to record — typed, written, drawn, spoken or annotated.
- If frustration rises, it is fine to pause and return later; the lesson holds their place.
- Avoid any language about age, school year or pace as a judgement of ability.

Recognising progress

Progress is described as Approaching Proficiency → Developing Proficiency → Proficiency → Refining Proficiency across understanding, application, communication and reflection. These describe where a learner is right now, not labels of what they are. Real understanding shown in part is still real understanding, even while other pieces are still settling.

Where they are	What it can look like
Approaching Proficiency	Beginning to engage; names some features and makes a start with support.

Developing Proficiency	Carries out the method with prompts; can explain part of the reasoning.
Proficiency	Works through independently and explains why each step is true.
Refining Proficiency	Applies the idea confidently to a new or unfamiliar case, and could teach it to someone else.

A note on evidence

Evidence can be captured in whatever form suits the learner — a photo of working, a short spoken explanation, an annotated Scratchpad image, or a written solution. There is no single required format. Incomplete work is information for the next connection, never a failure: it tells you exactly where to begin next time. The aim is a true record of the learner's thinking, not a tidy performance.

If a learner is stuck or distressed

Slow the pace and reduce the load: return to one concrete, familiar case and a single small question. Offer a Reconnection Route. Remind them they can pause and come back. A short, successful, low-stakes step rebuilds confidence more reliably than pushing through. Connection before curriculum — always.

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